

Lecture 17: Project

GRAPH RAG

Complete Notes (First-Thought Principle)

1 PROBLEM STATEMENT (START FROM REALITY)

We have **movie-related information** and we want to answer **complex questions** like:

- Movies directed by Christopher Nolan
- Actors who worked in Nolan movies
- Actors who won Oscar in a different movie and genre
- Movies similar to Inception

Constraints

- Data is **static** (facts already exist)
- Questions are **relational**
- We must **avoid hallucination**
- System should be **scalable**

So the problem is **NOT**:

| "Generate an answer"

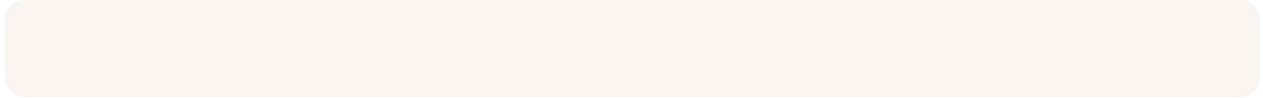
The problem **IS**:

| Retrieve correct facts and present them clearly

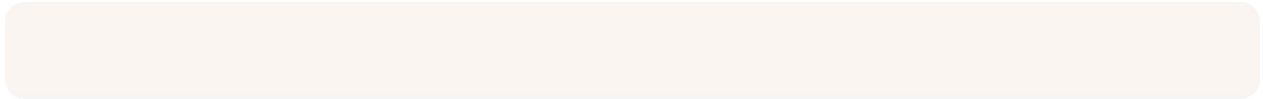
2 RAW DATA (MOVIE DOCUMENTATION EXAMPLE)

Before any database, imagine we only have **documents**.

Example: Movie Document – Inception



Example: Movie Document – Interstellar



Key Observation

Documents are:

- Repetitive
- Hard to connect
- Hard to query deeply

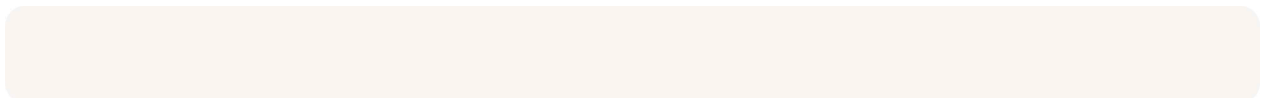
Example:

| “Which actors worked with Nolan and later won Oscar in another genre?”

This is **hard in documents**, but **natural in graphs**.

3 GRAPH THINKING (FIRST-THOUGHT PRINCIPLE)

Humans naturally think like:



This is NOT tabular thinking.

This is connected thinking.

4 GRAPH DATA MODEL

4.1 Node Types (CLASSES)

These are fixed.

4.2 Node Examples (INSTANCES)

- 🔑 Node label = class
 - 🔑 Property = actual data
-

4.3 Relationship Types (FACTS)

Relationships represent **truths**, not actions.

4.4 Example Graph Facts

5 WHAT GOES INTO VECTOR DATABASE?

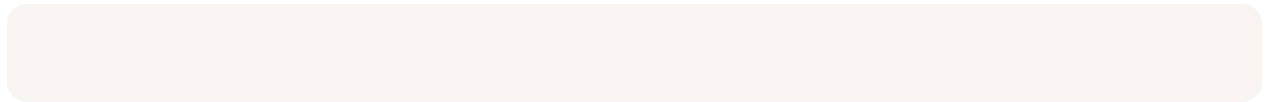
Vector DB does NOT store facts.

It stores semantic meaning of documents.

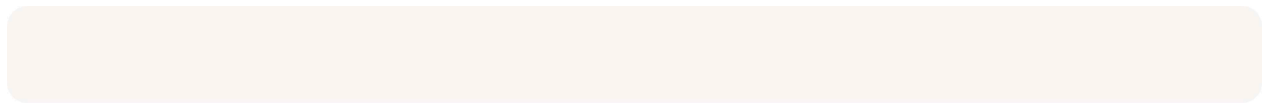
What is embedded?

Entire movie document text

Example text for Inception:



Vector DB Entry Example



What Vector DB is used for

- "Movies like Inception"
- "Mind-bending sci-fi movies"
- "Recommend similar films"

✗ Vector DB does NOT know:

- Director
- Actor
- Genre truth
- Awards

It only reduces search space.

6 INDEXING PHASE (VERY IMPORTANT)

Indexing happens once.

Step 1: Parse Documents

For each movie document:

- Extract entities
 - Extract relationships
 - Extract metadata
-

Step 2: Create Graph Nodes

Step 3: Create Relationships

Step 4: Store Embeddings in Vector DB

7 QUERYING PHASE (CORE LOGIC)

First Question We Ask

| Is this query FACTUAL or SIMILARITY-BASED?

7.1 FACTUAL QUERIES → GRAPH FIRST

Examples:

- Movies directed by Nolan
- Actors who won Oscar
- Movies that won Oscar in 2010

➡ Graph DB alone is enough

7.2 SIMILARITY QUERIES → VECTOR FIRST

Examples:

- Movies like Inception
- Recommend something similar
- Mind-bending sci-fi movies

➡ Vector DB → shortlist → Graph DB filter

8 DECISION FLOW (VERY IMPORTANT)

9 QUERY TEMPLATES (FIRST-THOUGHT PRINCIPLE)

We do **NOT** create templates per question.

We create **templates per thinking step**.

Alright. I'll do this **slowly, cleanly, and from zero**, exactly like **proper class notes**.

No shortcuts. No ego. No extra theory.

Only **TEMPLATES**, explained in **depth**, with **why, how, and examples**.

GRAPH QUERY TEMPLATES

First-Thought Principle (Detailed Notes)

These notes explain **how to think about queries**, not how to memorize Cypher.

0 What do we mean by “Template”?

A template is NOT a full query.

A template is NOT question-specific.

A template is:

A reusable thinking pattern that can be composed to answer any graph query.

Just like:

- loops & conditions in programming
 - not full programs
-

1 TEMPLATE 1: GRAPH PATTERN (Traversal)

First-thought principle

“Some things are connected to other things.”

That’s it.

This is the **core** of graph querying.

Mental model

If I say:

Nolan directed Inception

That sentence already has a pattern:

This is **one fact**.

Template shape (conceptual)

This is the **only structural operation** in graph queries.

Cypher form

This does NOT mean:

- search
- filter
- return

It only means:

“Match this connection if it exists.”

Example 1 (simple)

Meaning:

“A director who directed a movie.”

Example 2 (same template, different meaning)

Meaning:

“An actor who acted in a movie.”

Same template.

Only labels and relationship names change.

Very important rule

Repeating traversal does NOT create a new template.

This:



is just:

- traversal
- traversal again

Nothing new.

2 TEMPLATE 2: NODE PROPERTY ACCESS

First-thought principle

“Tell me something ABOUT this thing itself.”

This is **not** traversal.

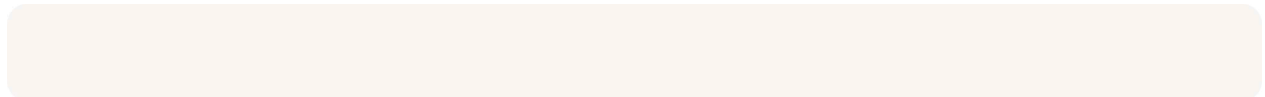
When this template is needed

Questions like:

- What is Nolan’s DOB?
- What year was Inception released?
- What is the name of this actor?

Here, you are **not** moving anywhere.

Template shape (conceptual)



Cypher form

Example

Meaning:

“Give me stored attributes of Nolan.”

Important distinction

Operation	Purpose
Traversal	Discover connections
Property access	Read stored data

Do not mix them.

3 TEMPLATE 3: FILTER (Constraint)

First-thought principle

“From what I already have, remove what I don’t want.”

Filtering **never creates data**.

Filtering **never moves the graph**.

Template shape (conceptual)

Cypher form

Example 1: Equality

Meaning:

| "Keep only Oscar awards."

Example 2: Inequality

Meaning:

| "Remove anything related to Nolan."

Example 3: Multiple filters (AND)

Meaning:

| "Apply filter, then apply another filter."

Critical rule

| AND / OR / NOT are NOT templates. They are filter logic.

4 TEMPLATE 4: SET COMBINATION (Result Control)

This is **not** traversal.

This is **result** management.

4.1 AND (Intersection)

First-thought:

“All conditions must be true.”

How it works:

- Same pattern
- Multiple filters

4.2 OR (Union)

First-thought:

“Either condition is acceptable.”

Two valid ways:

Way 1: OR inside filter

Way 2: UNION (cleaner for LLM systems)

4.3 NOT (Exclusion)

First-thought:

“Remove items matching this condition.”

or

Key rule

Set logic never changes graph structure. It only changes which matches survive.

5 TEMPLATE 5: PROJECTION (RETURN)

First-thought principle

“What exactly should I show to the user?”

Traversal & filtering may produce **large structures**.

Projection decides:

- what field
 - what shape
 - what uniqueness
-

Template shape (conceptual)

Cypher form

Examples

Return names only:

Return unique values:

Return count:

Important rule

RETURN does not affect logic. It only affects presentation.

6 How ALL templates work together (important)

Every graph query is just:

No exception.

7 Example (Full breakdown, no shortcut)

Question

Actors who worked in Nolan movies that won Oscar

Step-by-step thinking

1. Director connected to Movie
2. Movie connected to Award
3. Movie connected to Actor
4. Keep Oscar movies
5. Return actor names

Cypher using templates

